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THIRD REVIEW UNDER THE EXTENDED CREDIT FACILITY ARRANGEMENT AND REQUEST FOR WAIVER OF NONOBSERVANCE OF PERFORMANCE CRITERION AND MODIFICATION OF PERFORMANCE CRITERIA— DEBT SUSTAINABILITY ANALYSIS UPDATE[1]

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The updated Debt Sustainability Analysis indicates that Liberia's risk of external debt distress remains low. The present value of the external debt stock is projected to remain sustainable with all external debt indicators below the policy-related thresholds. Nonetheless, the pace of new borrowing has accelerated over the past year. This rapid accumulation of new loan commitments remains broadly consistent with the temporary scaling up of public investment envisaged under the program, but the authorities should continue to prioritize new financing for strategic projects on highly concessional terms to ensure that public debt remains sustainable.

[1] This document is the annual update of the analysis presented at the time of the current ECF request in November 2012 (IMF Country Report No. 12/340).

KEY ASSUMPTIONS UNDER THE BASELINE SCENARIO

1. **The Debt Sustainability Analysis (DSA) update indicates that Liberia continues to have a low risk of debt distress**. Although recent external debt accumulation, on a contractual basis, has been faster than initially envisaged, the pace of disbursements has been slower than anticipated. Compared with the last DSA update, the medium-term external debt profile (2014–17) is somewhat less favorable due to a relatively slower real GDP growth and a larger stock of restructured loans. However, the long-term debt profile is more favorable, with the improvement coming from higher nominal GDP after 2020. External debt would rise to 17.6 percent of GDP in FY2014, from 12.2 percent of GDP in FY2013, and would peak at 32.8 percent of GDP in FY2021, while in the previous update external debt would peak at 34.2 percent of GDP by FY2022. Public sector debt would rise from 13.7 percent of GDP in FY2013 to 20.1 percent of GDP in FY2014, peaking at 34.9 percent of GDP in FY2021 (compared with 39.3 percent in the previous update).

2. **The analysis reflects the impact of the following changes compared with the previous DSA update**.

- ***GDP growth and current account developments***. Compared with the December 2013 DSA update, the medium-term real GDP growth forecast has been revised slightly downward, mainly due to delays in mining activity, trends in world iron-ore prices and unresolved issues in the forestry sector. At the same time, higher projected inflation is raising nominal GDP.[2] In the long run, the effect of higher inflation on nominal GDP dominates. The current account deficit is projected to only gradually narrow over the projection period, reaching about 15 percent of GDP by 2030. It will continue to be sustainably financed by FDI and private flows associated with mining and forestry concessions.

- ***Restructured Loans***. **The Government of Liberia has completed the restructuring of pre-HIPC external debt with several creditors including** EIB/EU, ECOWAS, BADEA,[3] OFID, Kuwait, Saudi Arabia, ADB-NTF[4] and France. Overall the terms of the restructurings are less favorable than previously assumed. As a result, the end-June 2013 external debt stock (US$123 million, excluding loans from the IMF and Taiwan, Province of China) is US$26 million higher than envisaged in the previous DSA. There has been no progress on the restructuring of the Taiwanese loan.

[2] At the time of the 2nd ECF review staff lowered projected inflation over the long-run to reflect the lower utility costs associated with the execution of infrastructure projects in the energy sector. The projected inflation path has been revised upward to take into account real appreciation pressures from higher mining exports and to ensure consistency of long-run U.S. dollar inflation with the growth of the U.S. GDP deflator.

[3] BADEA: Arab Bank for Economic Development in Africa.

[4] ABD-NTF: African Development Bank-Nigeria Trust Fund.

- ***New external loan agreements***. After initial delays in securing external financing, the amount of newly-contracted external loan agreements has increased significantly since FY2013. The amount of external post-HIPC loan agreements ratified before the current ECF program (which started in November 2012) is US$142 million. From November 2012 to December 2013, excluding the ECF credit, the sum of ratified loan agreements amounts to US$377 million, with another US$228.3 million signed external loan agreements pending ratification. The amount of loans under negotiation is about US$476 million, of which US$416 million will be from IDA and the AfDB on highly concessional terms. In line with this rapid contracting of external loan agreements, the baseline macro-framework reflects higher borrowing and disbursements (US$3.5 billion, compared with US$2.9 billion in FY2014–FY2033 in the previous DSA update).

- ***Despite the rapid increase in contracted external loans, actual disbursements have been slow***. Excluding the ECF credit, only US$45 million (out of US$519 million ratified post-HIPC loan contracts) has been disbursed by the end of FY2013. Therefore, in the baseline scenario, although we assume that total disbursements will increase by about 20 percent between FY2014 and FY2033 compared with the amount assumed in the last DSA update, most of the increase is assumed to materialize after FY2023 (total disbursements between FY2014 and FY2022 are now assumed to be close to US$1.5 billion compared with US$1.4 billion in the previous DSA update).

PUBLIC AND EXTERNAL DEBT SUSTAINABILITY

3. **The external debt profile is less favorable than in the previous DSA update in the medium term but more favorable in the long term**. Compared with the last DSA update, due to slower real GDP growth and a larger stock of restructured loans, all external public and publicly-guaranteed (PPG) debt indicators (including the debt-to-GDP ratio) are now higher during 2014–17. The public sector debt-to-GDP ratio is also higher during the same period (Figures 1, 2 and Tables 1, 2). In the long term, however, despite the assumption of larger disbursements, the projected higher nominal GDP leads to more favorable external and public debt-to-GDP ratios.

4. **Stress tests indicate that external and public debt would remain sustainable, even under extreme scenarios, although the baseline forecast is subject to significant risks** (Figures 1, 2 and Tables 3, 4). On the upside, the baseline scenario only reflects the two mining projects currently under exploitation, although two additional major projects in their developmental phase are expected to come on stream around 2018. Furthermore, there is a potential for an upward revision of the GDP base,[5] which would lower debt-to-GDP ratios. On the downside, delays in the coming on stream of new mining projects could also lead to lower growth and government revenues. In the stress test of the new probability approach, Liberia's external public and publicly-guaranteed (PPG) debt-to-GDP ratio temporarily exceeds the threshold (Figure 4). However, given

[5] A revised set of national accounts for 2008–13 is expected to be published by the end of 2014.

that other debt distress measures are well below the thresholds, staff's view is that Liberia's external risk rating should remain "low". Public sector debt also continues to be sustainable, although the debt-to-GRP ratio and the debt-to-revenue ratio would increase in the long run under the scenario with a fixed primary balance (Figure 2 and Table 4).

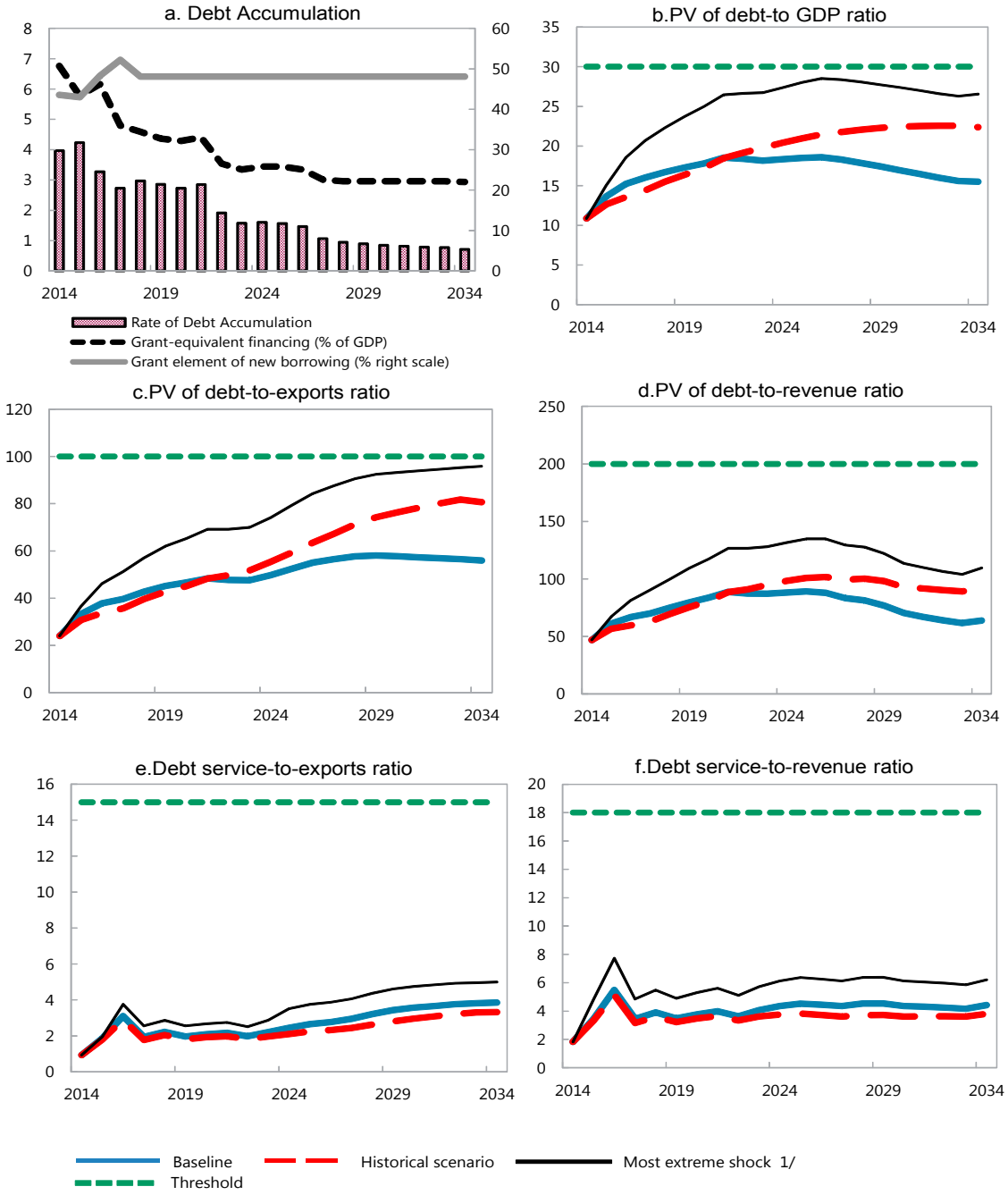
5. **While not an immediate concern for debt sustainability, the rapid pace of new borrowing might trigger a change in the external risk rating if it were to be sustained over the medium term**. In an alternative scenario, we assume that the government will sign another US$1.1 billion in new loan agreements (which is the same as the total amount of loan agreements that have been signed or in pipeline since the start of the current ECF program) from FY2016 to FY2018 so that the total amount of newly-signed external loans would be US$2.2 billion between FY2013 and FY2018. Whether this "sustained borrowing scenario" will result in a change of external risk rating depends on whether the faster contracting of loans would lead to an increase in total disbursements. In the baseline scenario, we assume that all loans signed before FY2015 will be disbursed before FY2021. Therefore, in the "sustained borrowing scenario" we assume all loans signed between FY2016 and FY2018 will be disbursed before FY2024 and that no other new loans will be signed from FY2019 to FY2024. This implies an increase of total disbursement by US$0.45 billion, compared with the baseline scenario. As shown in Figure 3, this will result in a change in the external risk rating from "low" to "medium".

6. **The authorities agreed with the assessment that external risk rating remains to be low**. They also emphasized that the large amount of new borrowings will finance the important infrastructure projects that contribute to addressing the country's binding growth constraints.

CONCLUSION

7. **The updated DSA shows that Liberia's debt profile remains sustainable under most scenarios**. Under the ECF-supported program, the government's borrowing plans are consistent with implementation of its poverty reduction strategy, the Agenda for Transformation. In particular, the government is actively seeking financing for strategic projects in the energy and transportation sectors in order to address the main binding constraints to broad-based economic growth. However, it must also be noted that, if the fast borrowing pace continues, a change in the external risk rating might be triggered. This highlights the need for the authorities to prioritize new financing for strategic projects and on highly concessional terms to ensure that public debt remains sustainable. Further strengthening project preparation, procurement, and monitoring would also help ensure that public investment is of high quality.

Figure 1. Liberia: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2014–34[1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Terms of borrowing shock; in c. to a Terms of borrowing shock; in d. to a Terms of borrowing shock; in e. to an Exports shock and in figure f. to a One-time depreciation shock. In the alternative scenarios of external DSA, historical average and standard deviation of major variables are calculated by using the data from 2008 due to the structual change of the economy.

Table 1. Liberia: External Debt Sustainability Framework, Baseline Scenario, 2011–34

(Percent of GDP, unless otherwise indicated)

	Actual			Historical 7/ Average	Standard 7/ Deviation	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**13.2**	**11.8**	**12.2**			**17.6**	**22.6**	**26.3**	**28.3**	**29.7**	**30.7**		**31.8**	**26.0**	
of which: public and publicly guaranteed (PPG)	13.2	11.8	12.2			17.6	22.6	26.3	28.3	29.7	30.7		31.8	26.0	
Change in external debt	2.0	-1.4	0.3			5.5	5.0	3.6	2.0	1.4	1.0		0.1	-0.1	
Identified net debt-creating flows	1.3	-0.4	1.4			3.5	4.1	3.2	1.9	1.1	1.0		-0.1	0.5	
Non-interest current account deficit	**36.8**	**31.7**	**31.4**	**27.2**	**12.1**	**40.8**	**40.5**	**27.8**	**22.6**	**28.8**	**30.1**		**16.7**	**15.2**	17.8
Deficit in balance of goods and services	95.6	84.5	72.1			72.0	66.4	48.8	39.4	40.9	39.3		19.6	17.4	
Exports	44.5	48.7	49.2			45.3	41.1	40.2	40.4	39.1	38.2		36.9	27.7	
Imports	140.1	133.2	121.3			117.3	107.5	89.0	79.8	80.0	77.5		56.4	45.1	
Net current transfers (negative = inflow)	-73.1	-66.8	-57.9	-102.7	36.4	-48.8	-41.1	-34.5	-29.5	-24.9	-21.1		-12.3	-8.1	-11.3
of which: official	-28.4	-26.8	-25.2			-24.2	-22.8	-21.0	-19.2	-17.3	-15.6		-10.9	-7.1	
Other current account flows (negative = net inflow)	14.3	14.0	17.2			17.6	15.2	13.6	12.7	12.9	11.9		9.4	5.9	
Net FDI (negative = inflow) 2/	**-34.1**	**-30.4**	**-28.8**	**-17.8**	**18.3**	**-36.5**	**-35.5**	**-23.4**	**-19.0**	**-25.6**	**-26.9**		**-15.6**	**-14.5**	-16.9
Endogenous debt dynamics 3/	**-1.5**	**-1.7**	**-1.2**			**-0.7**	**-0.8**	**-1.2**	**-1.7**	**-2.1**	**-2.2**		**-1.1**	**-0.1**	
Contribution from nominal interest rate	0.1	0.1	0.1			0.1	0.2	0.2	0.3	0.3	0.3		0.3	0.3	
Contribution from real GDP growth	-0.7	-0.9	-0.9			-0.8	-1.0	-1.4	-1.9	-2.4	-2.4		-1.5	-0.4	
Contribution from price and exchange rate changes	-0.9	-0.9	-0.4			...	...	...	...	...	...		...	...	
Residual (3-4) 4/	**0.7**	**-1.0**	**-1.1**			**2.0**	**0.9**	**0.4**	**0.1**	**0.3**	**0.0**		**0.2**	**-0.7**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 5/	...	...	7.7			10.9	13.7	15.2	16.0	16.7	17.3		18.3	15.5	
In percent of exports	...	...	15.6			24.0	33.2	37.8	39.6	42.8	45.2		49.7	55.9	
PV of PPG external debt	**...**	**...**	**7.7**			**10.9**	**13.7**	**15.2**	**16.0**	**16.7**	**17.3**		**18.3**	**15.5**	
In percent of exports	**...**	**...**	**15.6**			**24.0**	**33.2**	**37.8**	**39.6**	**42.8**	**45.2**		**49.7**	**55.9**	
In percent of government revenues	**...**	**...**	**27.8**			**47.0**	**61.4**	**66.8**	**70.0**	**74.9**	**79.7**		**88.2**	**63.9**	
Debt service-to-exports ratio (in percent)	**0.7**	**0.9**	**0.8**			**0.9**	**1.9**	**3.1**	**2.0**	**2.2**	**2.0**		**2.5**	**3.9**	
PPG debt service-to-exports ratio (in percent)	**0.7**	**0.9**	**0.8**			**0.9**	**1.9**	**3.1**	**2.0**	**2.2**	**2.0**		**2.5**	**3.9**	
PPG debt service-to-revenue ratio (in percent)	**1.3**	**1.6**	**1.4**			**1.8**	**3.4**	**5.5**	**3.4**	**3.9**	**3.5**		**4.4**	**4.4**	
Total gross financing need (Millions of U.S. dollars)	42.8	28.8	55.3			94.3	128.1	141.9	122.7	130.3	141.2		116.7	209.7	
Non-interest current account deficit that stabilizes debt ratio	34.8	33.1	31.1			35.3	35.4	24.2	20.6	27.4	29.1		16.6	15.3	
Key macroeconomic assumptions															
Real GDP growth (in percent)	7.0	8.1	8.5	7.4	1.9	7.3	6.4	7.0	8.3	9.6	9.3	8.0	5.0	1.5	5.2
GDP deflator in US dollar terms (change in percent)	8.8	7.0	3.6	6.6	2.4	3.1	3.6	4.0	3.7	3.6	3.5	3.6	2.9	3.7	3.2
Effective interest rate (percent) 6/	1.2	0.8	1.2	1.1	0.3	0.8	1.3	1.2	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Growth of exports of G&S (US dollar terms, in percent)	25.0	26.6	13.6	17.5	16.4	1.8	0.2	8.7	12.8	9.9	10.6	7.3	4.3	5.7	6.3
Growth of imports of G&S (US dollar terms, in percent)	15.6	10.0	2.4	16.8	25.4	6.9	1.1	-7.9	0.7	13.7	9.6	4.0	5.2	5.1	4.7
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	43.5	43.0	48.2	52.2	48.1	48.1	47.2	48.1	48.1	48.1
Government revenues (excluding grants, in percent of GDP)	23.7	26.3	27.7			23.1	22.3	22.8	22.9	22.3	21.7		20.8	24.2	22.4
Aid flows (in Millions of US dollars) 8/	40.3	28.3	45.7			80.5	59.6	70.0	56.0	63.6	71.9		119.5	245.5	
of which: Grants	40.3	28.3	45.7			80.5	59.6	70.0	56.0	63.6	71.9		119.5	245.5	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 9/	...	...	...			6.8	5.8	6.2	4.8	4.6	4.4		3.4	2.9	3.3
Grant-equivalent financing (in percent of external financing) 9/	...	...	...			65.0	58.3	63.1	65.2	62.2	63.1		68.9	74.4	70.9
Memorandum items:															
Nominal GDP (Millions of US dollars)	1414	1636	1840			2034	2242	2494	2801	3180	3596		5973	12276	
Nominal dollar GDP growth	16.5	15.7	12.4			10.5	10.2	11.2	12.3	13.5	13.1	11.8	8.0	5.2	8.5
PV of PPG external debt (in Millions of US dollars)			137.0			210.0	296.0	369.4	437.4	520.5	611.1		1077.9	1873.3	
(PVt-PVt-1)/GDPt-1 (in percent)						4.0	4.2	3.3	2.7	3.0	2.9	3.3	1.6	0.7	1.4
Gross workers' remittances (Millions of US dollars)	...	...	...			...	...	...	...	...	...		...	...	
PV of PPG external debt (in percent of GDP + remittances)	...	...	7.7			10.9	13.7	15.2	16.0	16.7	17.3		18.3	15.5	
PV of PPG external debt (in percent of exports + remittances)	...	...	15.6			24.0	33.2	37.8	39.6	42.8	45.2		49.7	55.9	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	0.8			0.9	1.9	3.1	2.0	2.2	2.0		2.5	3.9	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Includes private financing flows, including for iron-ore related investment which was included in FDI in the previous DSA

3/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

4/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

5/ Assumes that PV of private sector debt is equivalent to its face value.

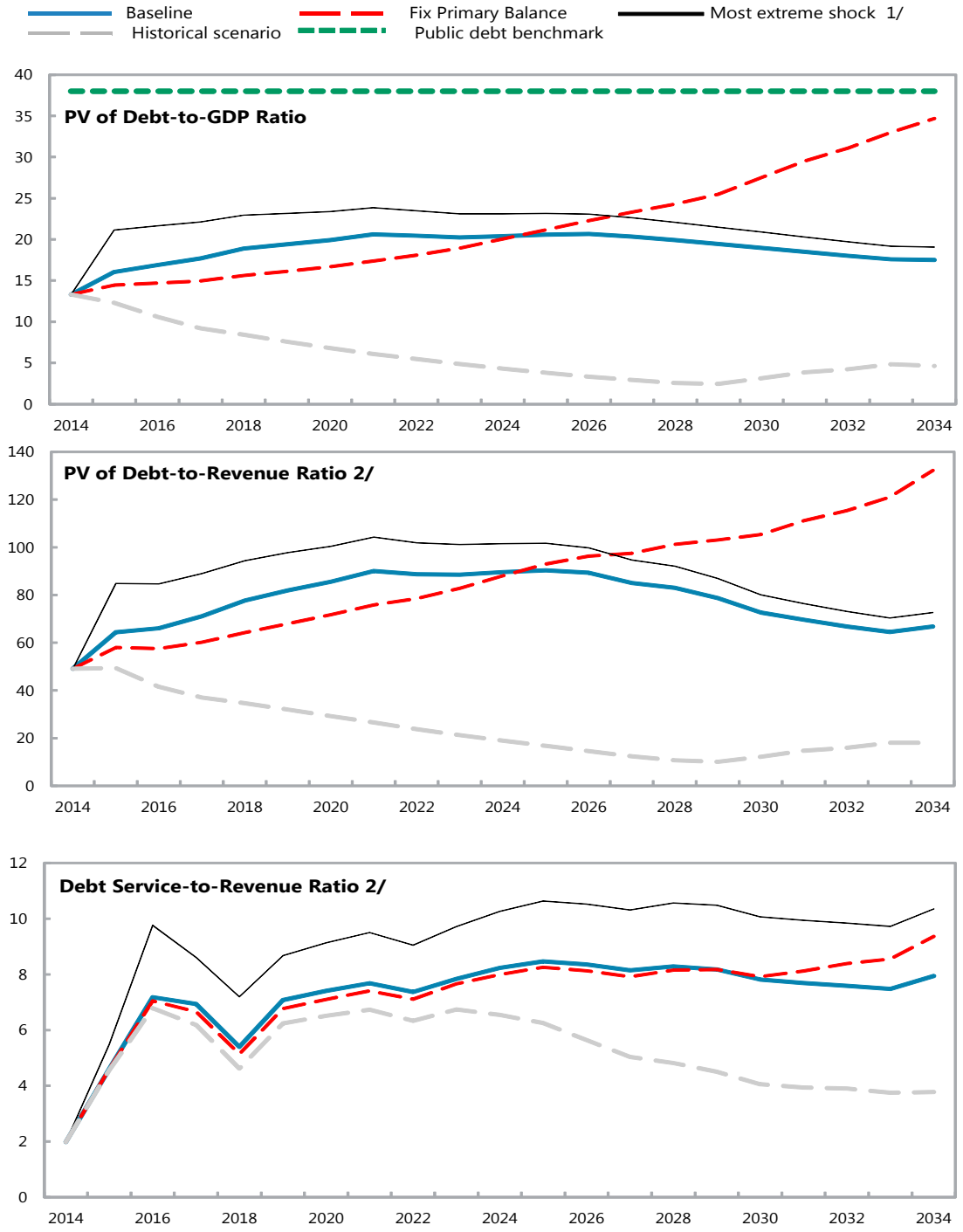
6/ Current-year interest payments divided by previous period debt stock.

7/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

8/ Defined as grants, concessional loans, and debt relief.

9/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Figure 2. Liberia: Indicators of Public Debt Under Alternative Scenarios, 2014–34[1]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

Table 2. Liberia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–34

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	15.1	13.4	13.7			20.1	25.0	28.0	30.0	31.9	32.8		33.8	28.0	
of which: foreign-currency denominated	15.1	13.4	13.7			20.1	25.0	28.0	29.0	30.0	30.9		31.9	26.1	
Change in public sector debt	1.5	-1.7	0.2			6.4	5.0	2.9	2.0	1.9	0.9		0.1	-0.1	
Identified debt-creating flows	-113.9	1.3	0.4			2.7	4.9	2.7	2.3	1.8	0.8		-0.2	-1.1	
Primary deficit	0.4	3.3	1.4	-0.4	2.0	3.6	6.7	5.0	5.1	5.1	4.2	5.0	1.9	-0.1	0.9
Revenue and grants	26.5	28.0	30.2			27.1	24.9	25.6	24.9	24.3	23.7		22.8	26.2	
of which: grants	2.8	1.7	2.5			4.0	2.7	2.8	2.0	2.0	2.0		2.0	2.0	
Primary (noninterest) expenditure	26.9	31.3	31.6			30.7	31.7	30.6	29.9	29.4	27.9		24.7	26.2	
Automatic debt dynamics	-2.0	-2.0	-1.0			-0.9	-1.8	-2.3	-2.8	-3.3	-3.4		-2.1	-1.1	
Contribution from interest rate/growth differential	-1.0	-1.3	-1.1			-0.9	-1.2	-1.7	-2.3	-2.8	-3.0		-1.8	-0.6	
of which: contribution from average real interest rate	-0.1	-0.2	-0.1			0.0	0.1	0.0	-0.2	-0.2	-0.2		-0.2	-0.2	
of which: contribution from real GDP growth	-0.9	-1.1	-1.1			-0.9	-1.2	-1.6	-2.1	-2.6	-2.7		-1.6	-0.4	
Contribution from real exchange rate depreciation	-1.0	-0.7	0.1			0.1	-0.7	-0.7	-0.4	-0.5	-0.5		...	...	
Other identified debt-creating flows	-112.4	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-112.4	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	115.4	-3.0	-0.2			3.7	0.1	0.3	-0.3	0.1	0.2		0.3	1.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	9.2			13.3	16.0	16.9	17.7	18.9	19.4		20.4	17.5	
of which: foreign-currency denominated	...	...	9.2			13.3	16.0	16.9	16.7	17.0	17.5		18.5	15.6	
of which: external	...	...	7.7			10.9	13.7	15.2	16.0	16.7	17.3		18.3	15.5	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	0.9	3.8	2.3			4.2	7.9	6.8	6.8	6.4	5.9		3.8	2.0	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	30.5			49.1	64.4	66.1	71.1	77.7	81.9		89.6	66.8	
PV of public sector debt-to-revenue ratio (in percent)	...	...	33.2			57.5	72.1	74.2	77.3	84.6	89.5		98.2	72.3	
of which: external 3/	...	...	27.8			47.0	61.4	66.8	70.0	74.9	79.7		88.2	63.9	
Debt service-to-revenue and grants ratio (in percent) 4/	1.6	1.7	3.0			2.0	4.6	7.2	6.9	5.4	7.1		8.2	7.9	
Debt service-to-revenue ratio (in percent) 4/	1.8	1.8	3.2			2.3	5.1	8.1	7.5	5.9	7.7		9.0	8.6	
Primary deficit that stabilizes the debt-to-GDP ratio	-1.1	5.0	1.2			-2.8	1.8	2.1	3.0	3.2	3.3		1.8	0.1	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.0	8.1	8.5	7.4	1.9	7.3	6.4	7.0	8.3	9.6	9.3	8.0	5.0	1.5	5.2
Average nominal interest rate on forex debt (in percent)	1.1	0.7	1.2	1.0	0.2	1.4	2.0	1.7	1.3	1.1	1.1	1.4	1.1	1.1	1.1
Average real interest rate on domestic debt (in percent)	-8.0	-5.2	-6.2	-6.5	1.4	...	...	...	...	4.9	1.3	3.1	1.3	0.4	1.0
Real exchange rate depreciation (in percent, + indicates depreciation)	-8.0	-5.1	1.0	-3.9	4.1	0.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	12.0	8.1	7.6	9.5	3.7	12.5	13.1	10.5	9.1	8.5	7.5	10.2	6.1	6.9	6.5
Growth of real primary spending (deflated by GDP deflator, in percent)	24.2	25.7	9.5	6.8	10.7	4.3	9.6	3.3	6.0	7.9	3.6	5.8	5.0	3.9	4.7
Grant element of new external borrowing (in percent)	...	...	...	...	...	43.5	43.0	48.2	52.2	48.1	48.1	47.2	48.1	48.1	...

Sources: Country authorities; and staff estimates and projections.

1/ It includes the gross debt contracted or guaranteed by the central government, excluding borrowing from the Central Bank of Liberia.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 3. Liberia: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34

(Percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	10.9	13.7	15.2	16.0	16.7	17.3	**18.3**	15.5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	10.9	12.6	13.6	14.4	15.5	16.4	**20.4**	22.4
A2. New public sector loans on less favorable terms in 2014-2034 2	10.9	15.0	18.6	20.7	22.3	23.7	**27.3**	26.5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	10.9	13.3	15.1	15.9	16.6	17.3	**18.3**	15.5
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	10.9	14.2	19.3	19.8	20.2	20.5	**20.4**	16.1
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	10.9	13.2	14.9	15.7	16.4	17.0	**18.1**	15.3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	10.9	6.4	-8.4	-5.9	-3.4	-1.2	**5.5**	10.7
B5. Combination of B1-B4 using one-half standard deviation shocks	10.9	-1.2	-21.9	-18.4	-14.9	-11.8	**-1.8**	7.8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	10.9	18.6	20.8	22.0	23.0	23.9	**25.4**	21.4
PV of debt-to-exports ratio								
Baseline	24	33	38	40	43	45	**50**	56
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	24	31	34	36	40	43	**55**	81
A2. New public sector loans on less favorable terms in 2014-2034 2	24	36	46	51	57	62	**74**	96
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	24	32	37	39	42	44	**49**	55
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	24	36	57	58	62	64	**66**	69
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	24	32	37	39	42	44	**49**	55
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	24	16	-21	-15	-9	-3	**15**	38
B5. Combination of B1-B4 using one-half standard deviation shocks	24	-3	-57	-48	-40	-32	**-5**	30
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	24	32	37	39	42	44	**49**	55
PV of debt-to-revenue ratio								
Baseline	47	61	67	70	75	80	**88**	64
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	47	57	60	63	69	75	**98**	92
A2. New public sector loans on less favorable terms in 2014-2034 2	47	67	82	90	100	109	**132**	110
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	47	60	66	69	75	80	**88**	64
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	47	64	85	86	90	94	**98**	67
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	47	59	65	69	74	79	**87**	63
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	47	29	-37	-26	-15	-6	**27**	44
B5. Combination of B1-B4 using one-half standard deviation shocks	47	-6	-96	-81	-67	-55	**-9**	32
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	47	83	92	96	103	110	**122**	88

Table 3. Liberia: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34 (Concluded)

(Percent)

Debt service-to-exports ratio								
Baseline	1	2	3	2	2	2	**2**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	1	2	3	2	2	2	**2**	3
A2. New public sector loans on less favorable terms in 2014-2034 2	1	2	2	2	3	3	**3**	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	1	2	3	2	2	2	**2**	4
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	1	2	4	3	3	3	**4**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	1	2	3	2	2	2	**2**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	1	2	3	1	1	1	**0**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	1	2	3	0	1	1	**-2**	1
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	1	2	3	2	2	2	**2**	4
Debt service-to-revenue ratio								
Baseline	2	3	5	3	4	3	**4**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	2	3	5	3	4	3	**4**	4
A2. New public sector loans on less favorable terms in 2014-2034 2	2	3	4	4	5	5	**6**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	2	3	6	4	4	4	**4**	4
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	2	3	6	4	4	4	**5**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	2	3	6	3	4	3	**4**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	2	3	5	2	2	2	**0**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	2	3	4	1	1	1	**-3**	1
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	2	5	8	5	5	5	**6**	6
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	46	46	46	46	46	46	**46**	46

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows. Historical averages and standard deviations used in external DSA stress tests are derived from post-2008 data due to structural changes of the economy.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

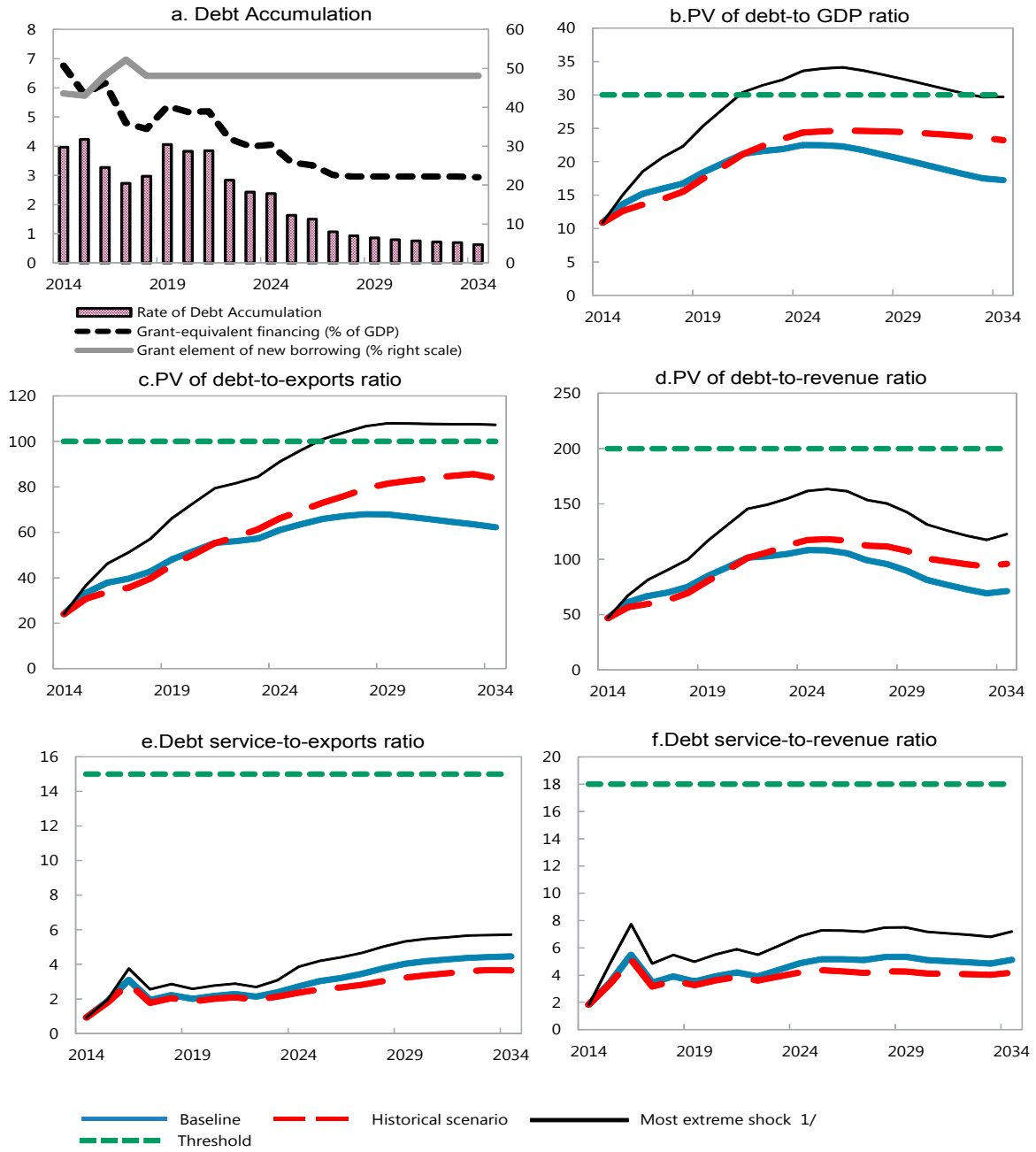
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 4. Liberia: Sensitivity Analysis for Key Indicators of Public Debt 2014–34

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	13	16	17	18	19	19	20	18
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	13	12	11	9	8	8	4	5
A2. Primary balance is unchanged from 2014	13	14	15	15	16	16	20	35
A3. Permanently lower GDP growth 1/	13	16	17	18	20	20	23	27
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	13	16	18	19	20	21	23	21
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	13	13	13	14	15	16	18	16
B3. Combination of B1-B2 using one half standard deviation shocks	13	13	12	13	15	16	18	17
B4. One-time 30 percent real depreciation in 2015	13	20	19	19	19	19	19	17
B5. 10 percent of GDP increase in other debt-creating flows in 2015	13	21	22	22	23	23	23	19
PV of Debt-to-Revenue Ratio 2/								
Baseline	49	64	66	71	78	82	90	67
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	49	49	41	37	35	32	19	18
A2. Primary balance is unchanged from 2014	49	58	58	60	64	68	88	132
A3. Permanently lower GDP growth 1/	49	65	67	73	81	86	102	103
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	49	65	69	75	83	88	100	82
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	49	54	50	56	63	69	79	62
B3. Combination of B1-B2 using one half standard deviation shocks	49	52	46	53	61	66	78	63
B4. One-time 30 percent real depreciation in 2015	49	82	76	76	79	80	83	65
B5. 10 percent of GDP increase in other debt-creating flows in 2015	49	85	85	89	94	98	102	73
Debt Service-to-Revenue Ratio 2/								
Baseline	2	5	7	7	5	7	8	8
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	2	5	7	6	5	6	7	4
A2. Primary balance is unchanged from 2014	2	5	7	7	5	7	8	9
A3. Permanently lower GDP growth 1/	2	5	7	7	6	7	9	9
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	2	5	7	7	6	7	9	9
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	2	5	7	6	5	7	8	8
B3. Combination of B1-B2 using one half standard deviation shocks	2	5	7	6	5	7	8	8
B4. One-time 30 percent real depreciation in 2015	2	5	10	9	7	9	10	10
B5. 10 percent of GDP increase in other debt-creating flows in 2015	2	5	8	8	6	7	9	8

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

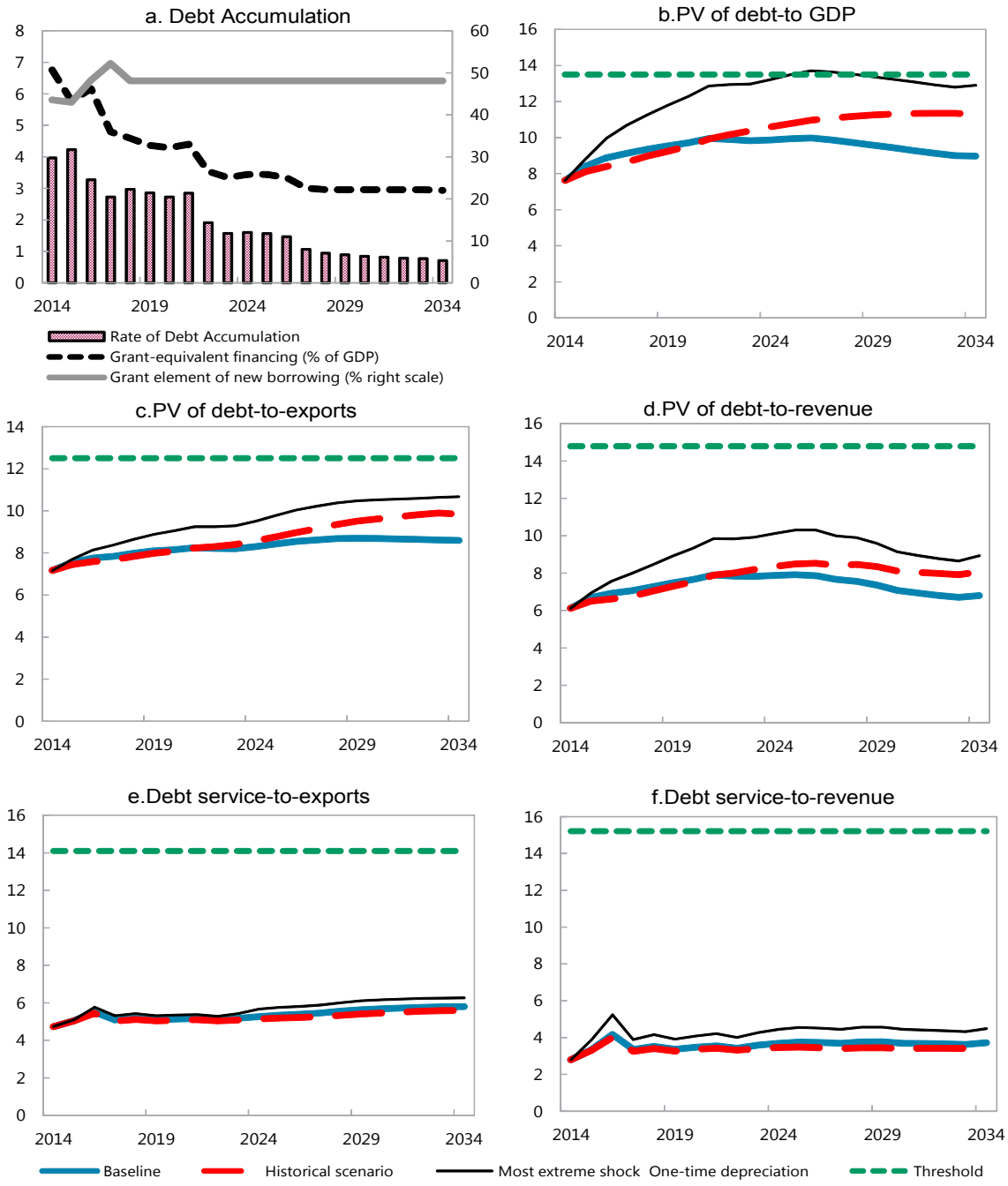
Figure 3. Liberia: Indicators of Public and Publicly Guaranteed External Debt Under Sustained Borrowing Scenarios, 2014–34[1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Terms of borrowing shock; in c. to a Terms of borrowing shock; in d. to a Terms of borrowing shock; in e. to an Exports shock and in figure f. to a One-time depreciation shock. In the alternative scenarios of external DSA, historical average and standard deviation of major variables are calculated by using the data from 2008 due to the structual change of the economy.

Figure 4. Liberia: Probability of Debt Distress of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2014–34[1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Terms of borrowing shock; in c. to a Terms of borrowing shock; in d. to a Terms of borrowing shock; in e. to an Exports shock and in figure f. to a One-time depreciation shock